

PRAJVAL ARORA

Robotics and Automation Engineering Student

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Robotics and Autonomous Systems sophomore seeking a Summer 2027 internship in robotics, automation, industrial robotics, or electromechanical engineering. Hands-on experience with ROS 2, NVIDIA Isaac Sim, Omniverse digital twins, PLC-integrated robot cells, machine vision, embedded hardware, control systems, systems integration, and physical AI.

EDUCATION

Arizona State University | B.S.E. Robotics and Autonomous Systems Engineering **Expected 2029**

- Focus: robotics, automation, controls, computer vision, embedded systems, robot kinematics, systems integration, industrial safety.

Harvard | Certificate of Internship and Entrepreneurship **Summer 2025**

TECHNICAL SKILLS

Robotics: ROS 2, NVIDIA Isaac Sim, Omniverse, digital twins, physical AI, robot kinematics, reinforcement learning, control systems, 6-DOF/4-DOF robots, UR5/UR3/URSe, cobots, JAKA, Dobot, Unitree Go2, FANUC

Automation: PLC programming, Allen-Bradley PLCs, structured text, SCADA, HMI, electrical panels, conveyors, industrial sensors, pneumatics, equipment commissioning, warehouse-management workflows, systems integration

Networks: Ethernet, USB, EtherCAT, digital I/O, IP-based communications, sensor and robot-cell network configuration

Mechanical: CAD/SolidWorks, GD&T, 3D printing, end-effector integration, inspection tools, electromechanical integration

Safety: Industrial safety, lockout/tag-out procedures, forklift awareness, robot-cell safety practices

Software: Python, C++, computer vision, image processing, AI agentic systems, local LLM inference, MATLAB, Simulink

TECHNICAL EXPERIENCE

Research Assistant | Arizona State University **Jul 2026–Present**

- Train robots in simulation with PhD researchers using ROS 2, Isaac Sim, Omniverse, digital twins, C++, Python, and RL workflows.
- Co-author an Isaac Sim lab manual for RAS 545, covering simulation, robot articulation, perception, and validation workflows.

Automation and PLC Co-Instructor Volunteer | TSMC Training Program **Jul 2026–Present**

- Train technicians on PLC programming, Allen-Bradley PLCs, structured text, sensors, conveyors, pneumatics, and robot-cell safety.
- Demonstrate troubleshooting across Ethernet, USB, EtherCAT, digital I/O, IP communications, and equipment commissioning.

Robotics Software and Electronics Lead | AMASS Lab, Arizona State University **Aug 2025–Mar 2026**

- Led ROS 2, embedded electronics, computer vision, and perception development for an autonomous blimp with live object detection.
- Integrated sensors, Jetson compute, electrical hardware, and control software for real-time autonomy and systems integration.

AR Assistant Developer | Barrow Institute, Arizona **Sept 2025–Mar 2026**

- Built AR speech therapy device with Jetson Nano LLM inference, stereo vision, birdbath AR display, and speech-guidance pipelines.
- Developed hardware integration, camera calibration, C++/Python prototyping, image processing, and embedded AI features.

Head of Research and Development | Tinker India Labs **Sept 2024–Jan 2025**

- Led robotics, IoT, CAD/SolidWorks prototyping, 3D printing, electronics assembly, and student-team mentoring for competitions.

SELECTED PROJECTS

Stellantis Industrial Robotics Demonstrator | ASU x Wandelbots **Aug 2026–Present**

- Developing a dual-URSe industrial automation demonstrator for the Stellantis Detroit event with Wandelbots.
- Hardware setup, end-effector integration, machine vision, network config, Isaac Sim digital-twin validation, and commissioning.

Automated Manufacturing Line Replica | Industrial Robot Cells [Innovation Showcase] **May 2025**

- Integrated UR5, UR3, JAKA, Dobot, FANUC concepts, cobots, PLCs, conveyors, sensors, vision, and pneumatics for a line replica.
- Applied PLC programming, digital I/O, Ethernet/IP networking, warehouse logic, safety procedures, and 6-DOF/4-DOF coordination.

Robotic Simulation and Reinforcement Learning | Unitree Go2 **Jul 2026–Present**

- Develop robot-training workflows in Isaac Sim with ROS 2 for Unitree Go2, exploring RL, digital twins, and sim-to-real pipelines.

Published Research and Microsoft Research Grant **Jun 2025**

- Awarded \$2,500 Microsoft Research Grant for dam-based cooling research; authored papers on sustainable data centers.

LEADERSHIP AND RECOGNITION

Treasurer, Marketing Head, and Software Project Lead | ARMS at ASU **2026–Present**

- Lead software and operations for MSN-affiliated projects; in talks with Phoenix Drone LLC for payload system under a \$10K budget.

Founder | FAITH Labs **Jun 2023–Present**

- Built school robotics labs in Kanpur, designed curriculum, taught 200+ students, coached teams to 1st regional and 10th national.

Founder and President | Tapri Builder Club **May 2025–Present**

- Built a startup team-formation platform reaching 22,000+ people, 2,200+ active members, and 40+ ventures launched.
- Honeywell Hackathon 2nd/3rd Place for turbo-jet heat-exchange systems; Robotics Championship 1st; BrainOBrain State Topper.